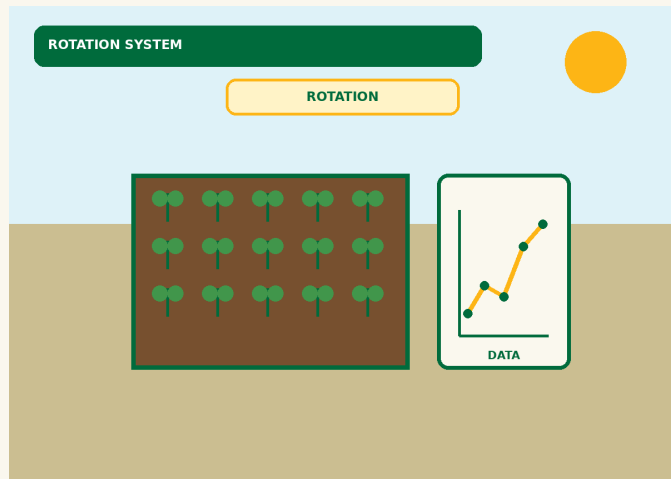




ROTATION

ROTATION SYSTEM

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 661 • ROTATION SYSTEM

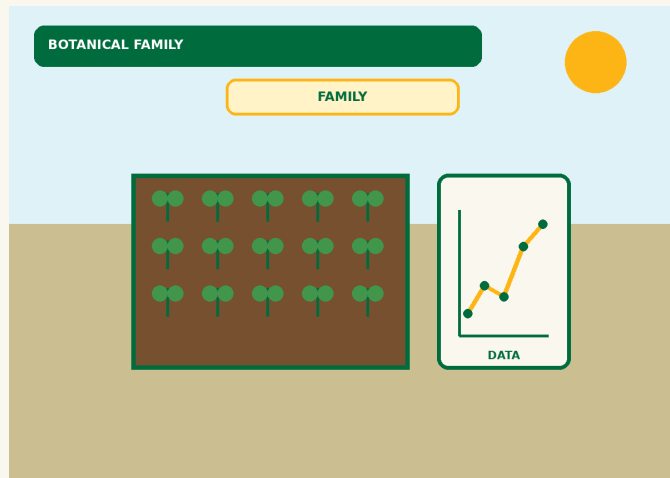
CONCEPT

A rotation system is a planned temporal sequence of crops designed around production, soil, pest, disease, and resource-management objectives.

INVESTIGATION TASK

Map the crop history of one bed.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



FAMILY

BOTANICAL FAMILY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 662 • BOTANICAL FAMILY

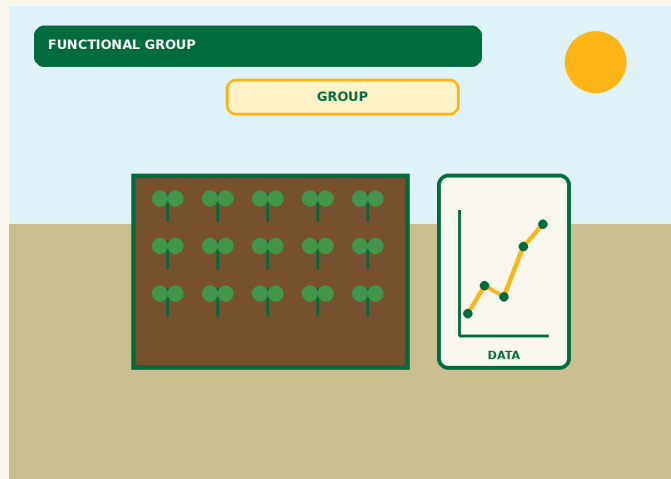
CONCEPT

Botanical family classification helps identify related crops that may share pests, diseases, and management characteristics.

INVESTIGATION TASK

Classify selected crops by family.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



GROUP

FUNCTIONAL GROUP

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 663 • FUNCTIONAL GROUP

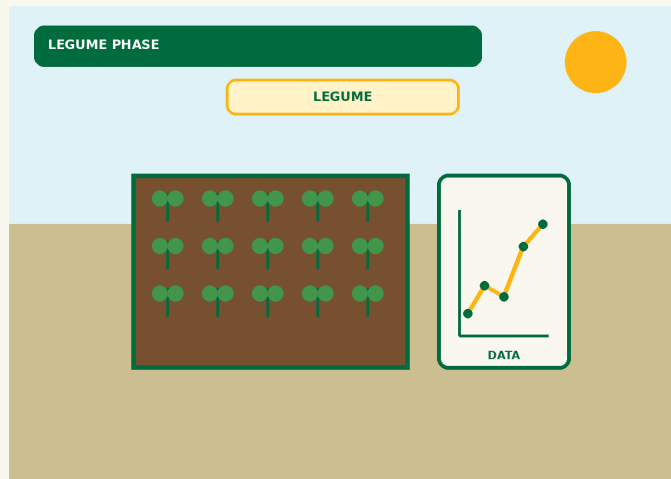
CONCEPT

Crops can also be grouped by production function, rooting pattern, harvested organ, growth habit, or nutrient demand.

INVESTIGATION TASK

Create two functional crop groups.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



LEGUME

LEGUME PHASE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 664 • LEGUME PHASE

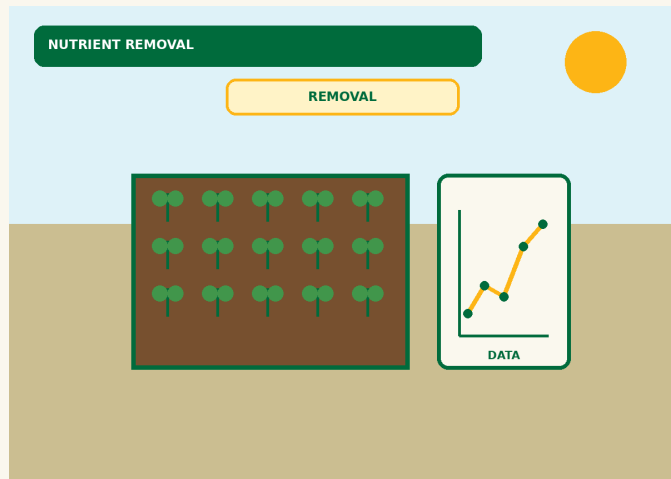
CONCEPT

A legume phase can influence nitrogen cycling through symbiosis with nitrogen-fixing bacteria.

INVESTIGATION TASK

Explain where a legume fits in the sequence.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



REMOVAL

NUTRIENT REMOVAL

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 665 • NUTRIENT REMOVAL

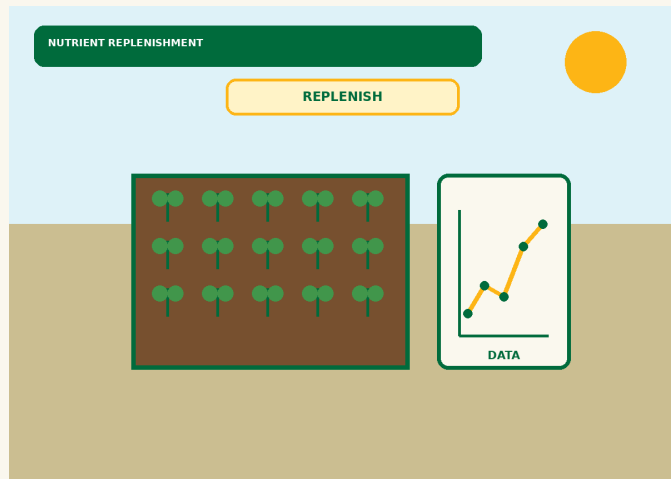
CONCEPT

Harvested crops remove nutrients from the production area in different quantities and proportions.

INVESTIGATION TASK

Compare teacher-provided removal data.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



REPLENISH

NUTRIENT REPLENISHMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 666 • NUTRIENT REPLENISHMENT

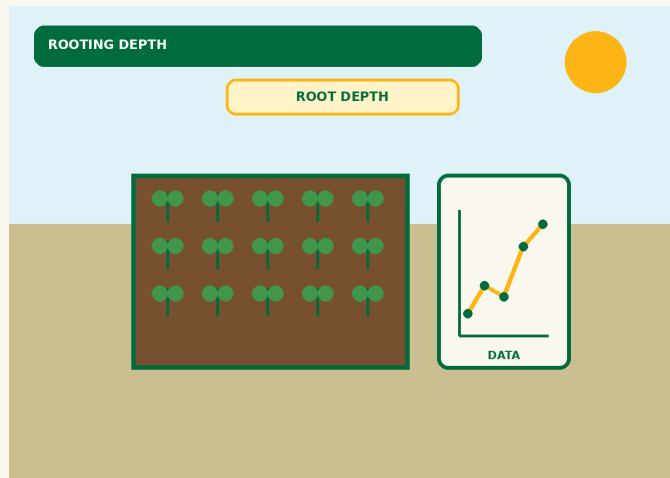
CONCEPT

Rotation planning can be coordinated with residues, compost, legumes, and nutrient additions to maintain fertility.

INVESTIGATION TASK

Propose a replenishment action after a demanding crop.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



ROOT DEPTH

ROOTING DEPTH

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 667 • ROOTING DEPTH

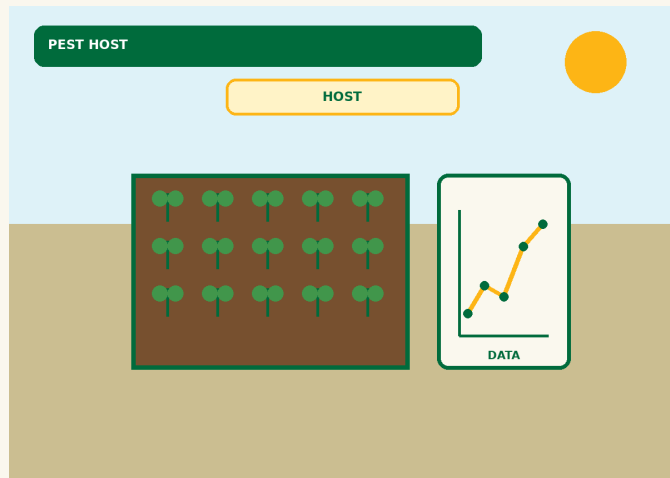
CONCEPT

Alternating crops with different rooting patterns can change how soil resources are explored.

INVESTIGATION TASK

Compare shallow- and deeper-rooted crops.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



HOST

PEST HOST

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 668 • PEST HOST

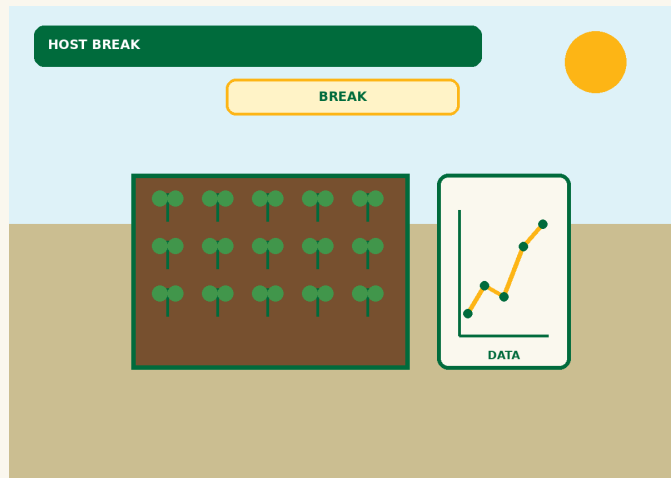
CONCEPT

A host crop supports the feeding or reproduction of a particular pest.

INVESTIGATION TASK

Identify related host crops in a scenario.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



BREAK

HOST BREAK

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 669 • HOST BREAK

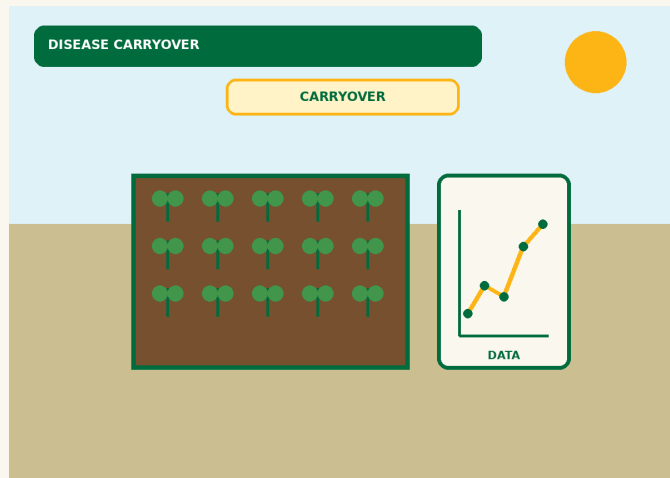
CONCEPT

A non-host crop can interrupt access to a preferred host and help reduce some pest cycles.

INVESTIGATION TASK

Choose a suitable break crop from provided options.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CARRYOVER

DISEASE CARRYOVER

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 670 • DISEASE CARRYOVER

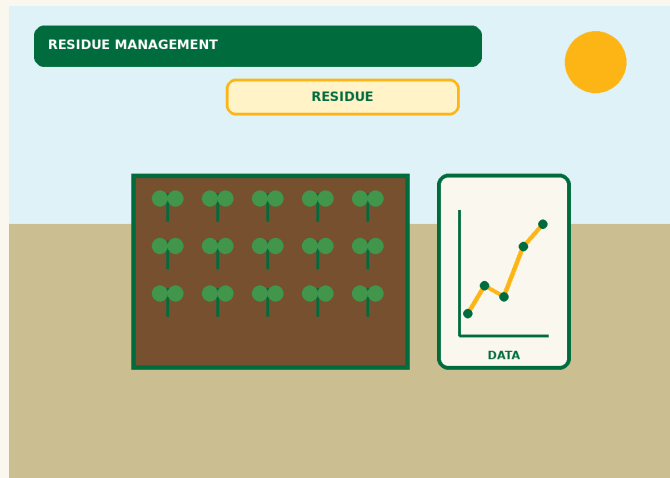
CONCEPT

Some pathogens persist in soil or residues and can affect susceptible crops in later cycles.

INVESTIGATION TASK

Use bed history to identify a disease-risk sequence.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RESIDUE

RESIDUE MANAGEMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 671 • RESIDUE MANAGEMENT

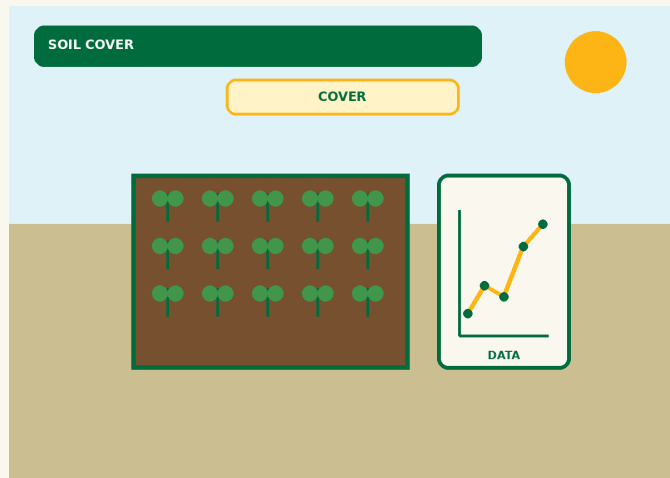
CONCEPT

Residue removal, incorporation, composting, or surface retention can affect soil condition and biological risks.

INVESTIGATION TASK

Evaluate two residue options.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



COVER

SOIL COVER

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 672 • SOIL COVER

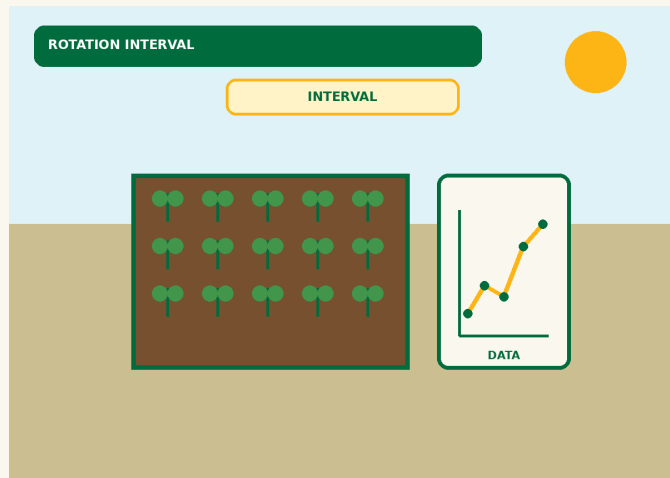
CONCEPT

Continuous or strategic soil cover can reduce erosion, suppress weeds, and influence soil moisture.

INVESTIGATION TASK

Calculate the proportion of cycles with cover.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



INTERVAL

ROTATION INTERVAL

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 673 • ROTATION INTERVAL

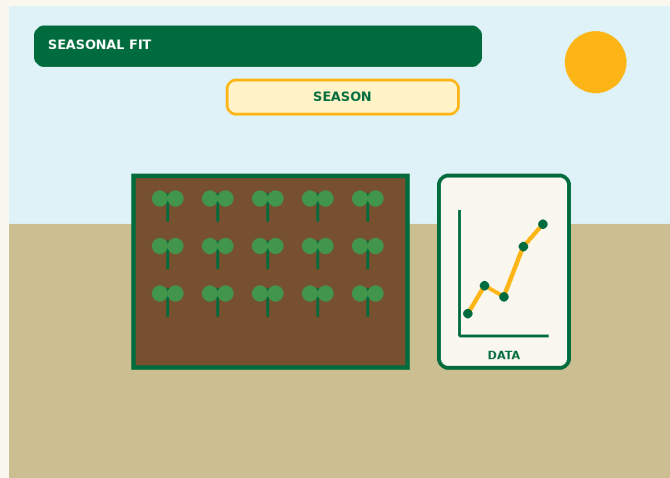
CONCEPT

Rotation interval is the time between susceptible crops or crop families returning to the same location.

INVESTIGATION TASK

Determine the interval in a four-cycle plan.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



SEASON

SEASONAL FIT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 674 • SEASONAL FIT

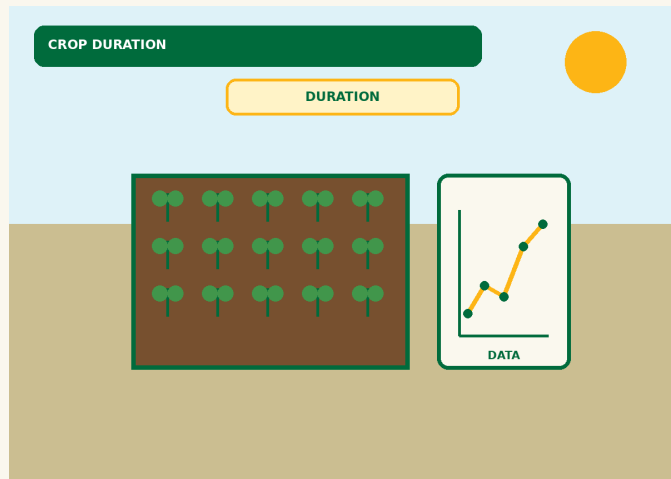
CONCEPT

A feasible rotation must match crop duration and requirements with seasonal production windows.

INVESTIGATION TASK

Check whether a proposed sequence fits the calendar.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



DURATION

CROP DURATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 675 • CROP DURATION

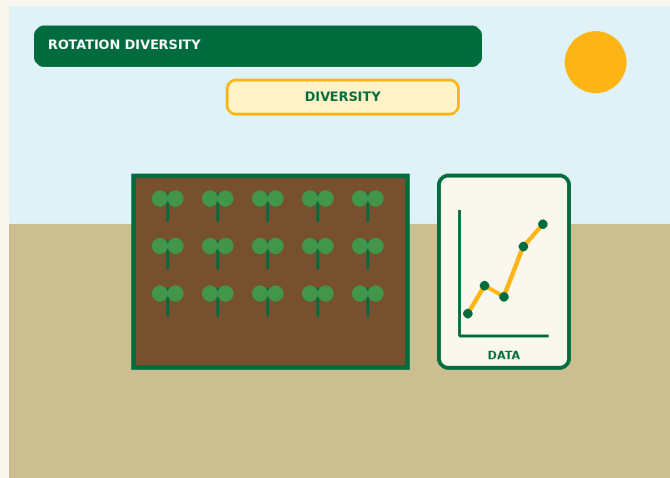
CONCEPT

Crop duration affects when land becomes available for the next crop and how many cycles can fit within a year.

INVESTIGATION TASK

Calculate a simple annual sequence.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



DIVERSITY

ROTATION DIVERSITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 676 • ROTATION DIVERSITY

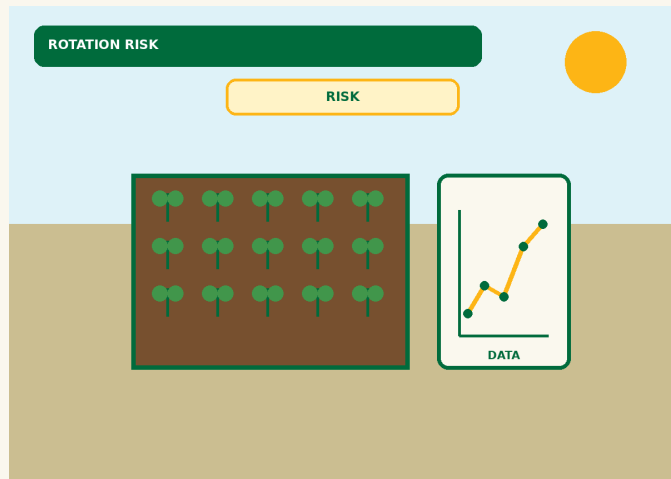
CONCEPT

Rotation diversity describes the variety of crop families, functions, and growth patterns represented through time.

INVESTIGATION TASK

Compare diversity in two plans.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RISK

ROTATION RISK

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 677 • ROTATION RISK

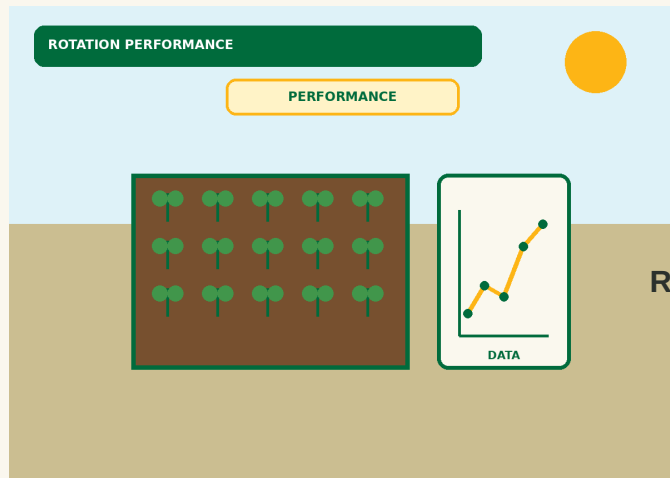
CONCEPT

Rotation risk includes pest, disease, nutrient, market, water, and timing constraints that can accumulate in a sequence.

INVESTIGATION TASK

Score two plans using a simple risk matrix.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PERFORMANCE

ROTATION PERFORMANCE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 678 • ROTATION PERFORMANCE

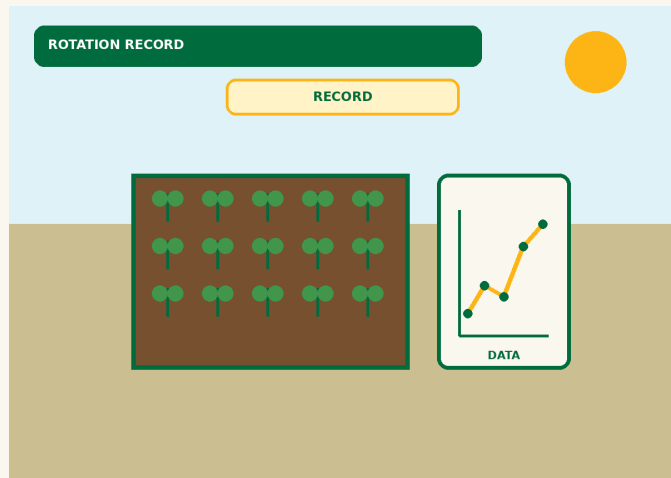
CONCEPT

Performance can be evaluated using yield, soil cover, pest incidence, input use, and production continuity.

INVESTIGATION TASK

Select three indicators to compare rotations.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RECORD

ROTATION RECORD

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 679 • ROTATION RECORD

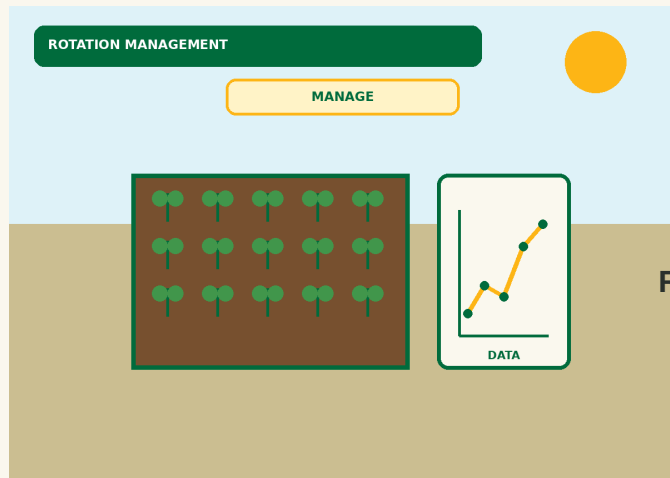
CONCEPT

Detailed rotation records connect crop identity, family, dates, inputs, residues, pests, yield, and next-crop decisions.

INVESTIGATION TASK

Audit one bed-history record.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MANAGE

ROTATION MANAGEMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 680 • ROTATION MANAGEMENT

CONCEPT

Rotation management uses historical evidence and production goals to select the next crop and associated practices.

INVESTIGATION TASK

Recommend a next crop with technical justification.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE